Problem

The area of a rectangle is $(6x + 9)\text{cm}^2$. Use factorisation to establish possible expressions for the dimensions of this rectangle.

Solution

When we factorise a linear expression such as $6x + 9$, we find the highest common factor of each term. In this case, that's 3, so 3 goes on the outside of the brackets. We then divide each term by this number and we are left with:

$$6x + 9 = 3(2x + 3)$$

The dimensions could be

3cm and $(2x + 3)\text{cm}$.

Problem

Two integers have a product of $2n^2 + 10n$. Use factorisation to find possible values of these numbers in terms of n.

Solution

When we factorise an expression such as $2n^2 + 10n$, we find the highest common factor of each term. The highest common factor of $2n^2$ and $10n$ is $2n$. We then divide each term by $2n$ to get:

$$2n^2 + 10n = 2n(n + 5)$$

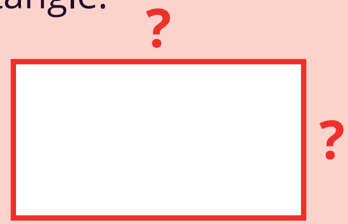
The two numbers could be

$2n$ and $n + 5$

(where n is an integer).

Problem

The area of a rectangle is $(14b^2 - 35b)\text{cm}^2$.
Use factorisation to establish possible expressions for the dimensions of this rectangle.



Solution

When we factorise an expression such as $14b^2 - 35b$, we find the highest common factor of each term.

The highest common factor of 14 and 35 is 7, so the highest common factor of $14b^2 - 35b$ is therefore $7b$. If we divide each term by $7b$, we are left with:

$$14b^2 - 35b = 7b(2b - 5)$$

The dimensions could be $7b$cm and $(2b - 5)$cm.

Problem

The area of a rectangle is $(x^2 - 3x - 28)\text{cm}^2$.
Use factorisation to establish possible expressions for the dimensions of this rectangle.

Work out the restrictions on the value of x, given these dimensions.

Solution

When we factorise a quadratic expression of the form $x^2 + bx + c$, we look for two numbers whose product is c and whose sum is b. In the case of $x^2 - 3x - 28$, we want two numbers whose product is −28 and whose sum is −3. These numbers are −7 and 4.

$$x^2 - 3x - 28 = (x - 7)(x + 4)$$

The dimensions could be $(x - 7)$cm and $(x + 4)$cm.

The dimensions of the rectangle must be greater than zero, so we form and solve an inequality with each length.

$x - 7 > 0$ or $x + 4 > 0$

$x > 7$ and $x > -4$

x must be greater than -4.

BEYOND MATHS

Factorising **Word Problems**

Problem

The product of two numbers is $x^2 - 10x + 24$. Use factorisation to work out two possible expressions for these numbers in terms of x.

Solution

When we factorise a quadratic expression of the form $x^2 + bx + c$, we look for two numbers whose product is c and whose sum is b.

In this question, we are looking for two numbers whose product is 24 and whose sum is −10. These numbers must be −6 and −4.

$$x^2 - 10x + 24 = (x - 6)(x - 4)$$

The two numbers could be $x - 6$ and $x - 4$.

Problem

The area of a square is $(x^2 - 12x + 36)\text{cm}^2$. Use factorisation to find expressions for the width of this square in terms of x.

Solution

When we factorise a quadratic expression of the form $x^2 + bx + c$, we look for two numbers whose product is c and whose sum is b.

In this question, we are looking for two numbers whose product is 36 and whose sum is −12. These numbers are −6 and −6.

$$\begin{aligned} x^2 - 12x + 36 &= (x - 6)(x - 6) \\ &= (x - 6)^2 \end{aligned}$$

The square has a width of $(x - 6)$cm.